

Problems on List Comprehension

Each problem includes a sample input and expected output for better understanding.

SECTION A - BASIC

Core syntax, built-in operations, and straightforward practice.

1. Write a Python program to create a list of squares using list comprehension.

Input	numbers = [1, 2, 3, 4, 5]
Expected Output	[1, 4, 9, 16, 25]

2. Write a Python program to extract all even numbers from a list using list comprehension.

Input	numbers = [1, 2, 3, 4, 5, 6, 7, 8]
Expected Output	[2, 4, 6, 8]

3. Write a Python program to extract all odd numbers from a list using list comprehension.

Input	numbers = [1, 2, 3, 4, 5, 6, 7]
Expected Output	[1, 3, 5, 7]

4. Write a Python program to convert all words in a list to uppercase using list comprehension.

Input	words = ['python', 'java', 'sql']
Expected Output	['PYTHON', 'JAVA', 'SQL']

5. Write a Python program to create a list containing the length of each word.

Input	words = ['AI', 'Python', 'LihaTech']
Expected Output	[2, 6, 8]

6. Write a Python program to select words having more than four characters.

Input	words = ['data', 'python', 'AI', 'model', 'cloud']
Expected Output	['python', 'model', 'cloud']

7. Write a Python program to double every number in a list using list comprehension.

Input	numbers = [2, 4, 6, 8]
Expected Output	[4, 8, 12, 16]

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8. Write a Python program to extract only positive numbers from a list.

Input	numbers = [-5, 3, -1, 0, 7, -2, 9]
Expected Output	[3, 7, 9]

9. Write a Python program to convert a string into a list of its characters using list comprehension.

Input	text = Python
Expected Output	['P', 'y', 't', 'h', 'o', 'n']

10. Write a Python program to extract only vowels from a string.

Input	text = LihaTech Kolhapur
Expected Output	['i', 'a', 'e', 'o', 'a', 'u']

11. Write a Python program to replace every negative number in a list with zero.

Input	numbers = [4, -2, 7, -5, 3]
Expected Output	[4, 0, 7, 0, 3]

12. Write a Python program to create a list containing the first character of each word.

Input	words = ['Python', 'Machine', 'Learning']
Expected Output	['P', 'M', 'L']

13. Write a Python program to create pairs of each number and its square.

Input	numbers = [1, 2, 3, 4]
Expected Output	[(1, 1), (2, 4), (3, 9), (4, 16)]

14. Write a Python program to flatten a two-dimensional list using list comprehension.

Input	matrix = [[1, 2], [3, 4], [5, 6]]
Expected Output	[1, 2, 3, 4, 5, 6]

15. Write a Python program to generate all multiples of 3 from 1 to 20 using list comprehension.

Input	Range = 1 to 20
Expected Output	[3, 6, 9, 12, 15, 18]

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16. Write a Python program to convert a list of numeric strings into integers.

Input	values = ['10', '20', '30', '40']
Expected Output	[10, 20, 30, 40]

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SECTION B - INTERMEDIATE

Multi-step logic and practical manipulation of the topic.

1. Write a Python program to label each number as Even or Odd using conditional list comprehension.

Input	numbers = [1, 2, 3, 4, 5]
Expected Output	['Odd', 'Even', 'Odd', 'Even', 'Odd']

2. Write a Python program to transpose a matrix using nested list comprehension.

Input	matrix = [[1, 2, 3], [4, 5, 6]]
Expected Output	[[1, 4], [2, 5], [3, 6]]

3. Write a Python program to find common elements between two lists using list comprehension.

Input	list1 = [1, 2, 3, 4, 5] list2 = [3, 4, 5, 6, 7]
Expected Output	[3, 4, 5]

4. Write a Python program to remove empty or whitespace-only strings from a list.

Input	values = ['Python', '', ' ', 'AI', 'Data']
Expected Output	['Python', 'AI', 'Data']

5. Write a Python program to flatten a nested list but keep only even numbers.

Input	matrix = [[1, 2, 3], [4, 5, 6], [7, 8]]
Expected Output	[2, 4, 6, 8]

6. Write a Python program to extract the names of students whose score is 75 or above.

Input	students = [('Asha', 82), ('Ravi', 68), ('Neha', 91), ('Aman', 75)]
Expected Output	['Asha', 'Neha', 'Aman']

7. Write a Python program to generate the Cartesian product of two lists.

Input	list1 = [1, 2] list2 = ['A', 'B']
Expected Output	[(1, 'A'), (1, 'B'), (2, 'A'), (2, 'B')]

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8. Write a Python program to generate all prime numbers between 2 and 30 using list comprehension.

Input	Range = 2 to 30
Expected Output	[2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

9. Write a Python program to extract all palindrome words from a list.

Input	words = ['level', 'python', 'madam', 'data', 'radar']
Expected Output	['level', 'madam', 'radar']

10. Write a Python program to extract the main diagonal elements of a square matrix.

Input	matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
Expected Output	[1, 5, 9]

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SECTION C - ADVANCED

Algorithmic problem solving and deeper application of the topic.

1. Write a Python program to find all unique elements that occur more than once while preserving their first appearance order.

Input	numbers = [4, 2, 4, 3, 2, 5, 4, 3]
Expected Output	[4, 2, 3]

2. Write a Python program to generate all pairs whose sum is equal to a given target.

Input	numbers = [1, 2, 3, 4, 5, 6] target = 7
Expected Output	[(1, 6), (2, 5), (3, 4)]

3. Write a Python program to create the transpose of a 3 x 3 matrix using nested list comprehension.

Input	matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
Expected Output	[[1, 4, 7], [2, 5, 8], [3, 6, 9]]

4. Write a Python program to flatten a three-level nested list into a single list.

Input	data = [[[1, 2], [3]], [[4, 5], [6]]]
Expected Output	[1, 2, 3, 4, 5, 6]

5. Write a Python program to calculate the sum of each row of a matrix using list comprehension.

Input	matrix = [[2, 4, 6], [1, 3, 5], [10, 20, 30]]
Expected Output	[12, 9, 60]

6. Write a Python program to assign grades to marks using a conditional list comprehension: A for 90+, B for 75-89, C for 60-74, and D below 60.

Input	marks = [95, 82, 67, 54, 90]
Expected Output	['A', 'B', 'C', 'D', 'A']

7. Write a Python program to convert comma-separated student records into tuples.

Input	records = ['Asha,85', 'Ravi,72', 'Neha,91']
Expected Output	[('Asha', 85), ('Ravi', 72), ('Neha', 91)]

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8. Write a Python program to create a multiplication table from 1 to 5 as a nested list.

Input	Range = 1 to 5
Expected Output	[[1, 2, 3, 4, 5], [2, 4, 6, 8, 10], [3, 6, 9, 12, 15], [4, 8, 12, 16, 20], [5, 10, 15, 20, 25]]

9. Write a Python program to generate all unique Pythagorean triples where $1 \leq a < b < c \leq 20$.

Input	Range = 1 to 20
Expected Output	[(3, 4, 5), (5, 12, 13), (6, 8, 10), (8, 15, 17), (9, 12, 15), (12, 16, 20)]

10. Write a Python program to generate sliding windows of size 3 from a list using list comprehension.

Input	numbers = [10, 20, 30, 40, 50, 60]
Expected Output	[[10, 20, 30], [20, 30, 40], [30, 40, 50], [40, 50, 60]]